PawLog Design Document

Nov 5, 2026 · @Eleah Burman

Project Description

PawLog is a web app that helps pet owners track their pets' daily health and catch problems early. Owners log meals, water, activity, weight, and symptoms each day, and a machine learning model flags entries that look off for that specific pet.

Pet health problems often start small: a dog eats a little less for a few days, or a cat drinks more water than usual. Owners tend to notice only once it's serious. PawLog keeps a daily record and compares each day to the pet's own normal, so owners get an early warning and a clear history to show the vet.

The app has two independent features:

- **Daily health log with ML health alerts** (Eleah): owners create, edit, delete, and filter daily entries. Each entry gets a probability that it needs attention, and owners can mark alerts as a real problem or a false alarm.
- **Pet profiles with a care schedule** (Bryan): owners manage pet profiles and care tasks like meds, vaccines, and vet appointments, with a dashboard of upcoming and overdue tasks.

Machine learning: a logistic regression model, written from scratch in JavaScript and trained with gradient descent. It uses 10 features per entry: how far food, water, activity, and weight are from the pet's 14-day normal, five symptom flags, and the number of low-food days in a row. It's trained on 1,500 labeled synthetic entries with a train/test split, and it retrains on user feedback. Alerts are not veterinary advice.

Tech stack: React with hooks, Node.js, Express, MongoDB (native driver), and Passport for login.

User Personas

Persona	Who they are	What they need	Main feature
Priya	Busy grad student with a dog and a cat. Travels sometimes and uses a pet sitter.	A clear health history she can pull up for her vet or sitter, and one place to see what's due across both pets.	Daily health log, care dashboard
Marcus	First-time cat owner. Doesn't know what's normal for his cat.	To know when something might be wrong before it gets serious.	ML health alerts
Dana	Has an older dog on daily meds.	To track med times, vaccines, and vet visits so nothing gets missed.	Pet profiles, care schedule

User Stories

Eleah: Daily Health Log and ML Health Alerts

1. Logging a day. As Priya, I want to create, edit, and delete daily health entries for each pet so that I have an accurate record of how they're doing.

Every night after dinner, Priya opens PawLog and logs that her dog Luna ate 2 cups, drank 3 cups of water, and walked 45 minutes. The next morning she realizes she typed the wrong weight, so she edits the entry. She deletes a duplicate she made by accident.

2. Showing the vet a history. As Priya, I want to view past entries and filter them by pet, date range, or symptom so that I can show the vet a clear history.

At Luna's checkup, the vet asks how long the coughing has been going on. Priya filters her entries to Luna, the last month, and "coughing." She shows the vet that it started 9 days ago and has come up 4 times since.

3. Getting an early warning. As Marcus, I want the app to tell me how likely it is that my cat's entry needs attention so that I can catch health problems early.

Marcus's cat Milo usually eats about a cup a day. For three days he's eaten half that. When Marcus logs the third day, PawLog shows an alert: Milo has eaten much less than usual 3 days in a row, with an 82% chance this needs attention. Marcus books a vet visit that

afternoon.

4. Teaching the model. As Marcus, I want to mark an alert as a real problem or a false alarm so that the predictions get more accurate for my cat over time.

A week later, PawLog flags a day when Milo was less active. Marcus knows it was just a lazy rainy day, so he marks it as a false alarm. The model retrains on his feedback and gets better at telling a real problem from an off day.

Bryan: Pet Profiles and Care Schedule

5. Setting up a profile. As Dana, I want to create, edit, and delete pet profiles with name, species, breed, age, and vet info so that all my pet's details are in one place.

Dana adopts Rocky, a 12-year-old beagle. She creates his profile with his breed, age, and her vet's phone number. When Rocky's birthday comes, she updates his age.

6. Never missing a med. As Dana, I want to create, edit, and delete care tasks like meds, vaccines, and vet appointments with dates and repeat schedules so that I never miss one.

Rocky takes heart medication every morning and gets a rabies booster every year. Dana adds both as repeating care tasks. When the vet changes his med time, she edits the task.

7. Seeing what's due. As Priya, I want to see a list of upcoming and overdue care tasks across all my pets so that I know what needs to be done this week.

Before a work trip, Priya opens the care dashboard. It shows that Luna's flea treatment is overdue and her cat's vaccine is due Friday. She handles the flea treatment that night and leaves a note for her sitter about the vaccine.

Design Mockups

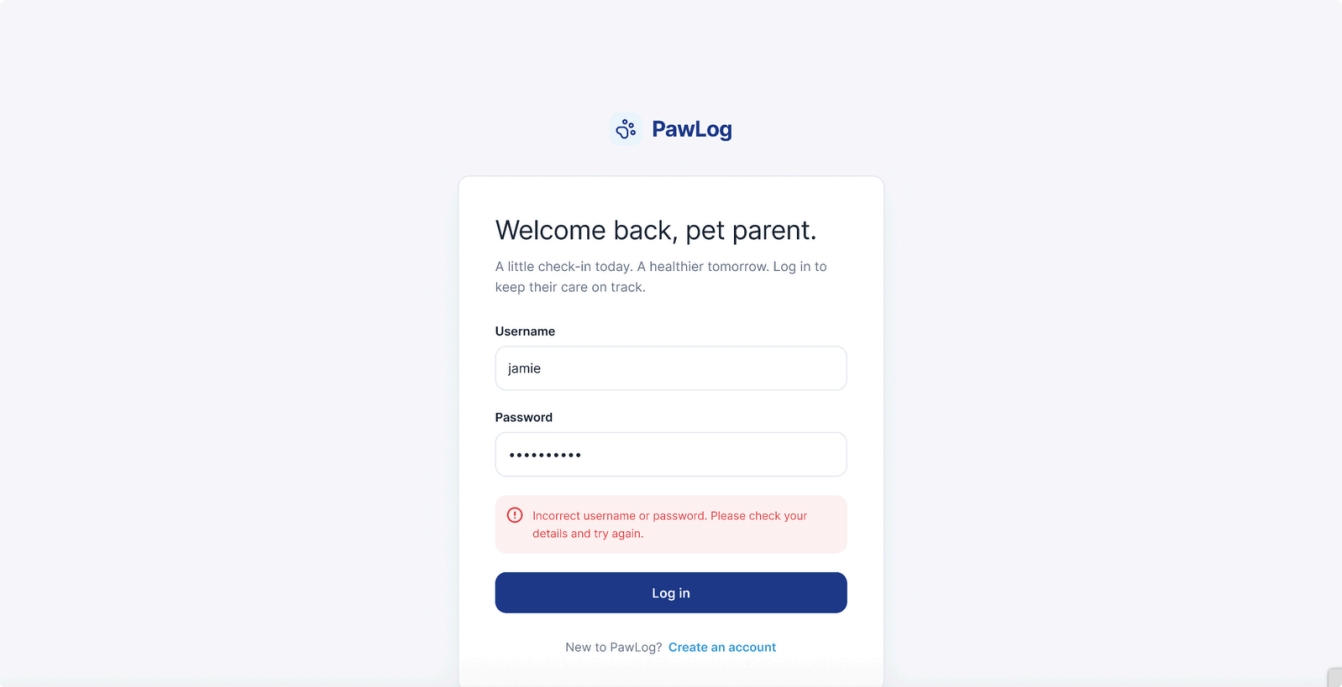
PawLog has 9 screens. The mockups were made in Figma and follow the table's order.

Screen	Owner	What it shows	Story
Login / Register	Eleah	Username and password fields, a Log in button, and a link to switch to Register. Error message area for wrong passwords or taken usernames.	All

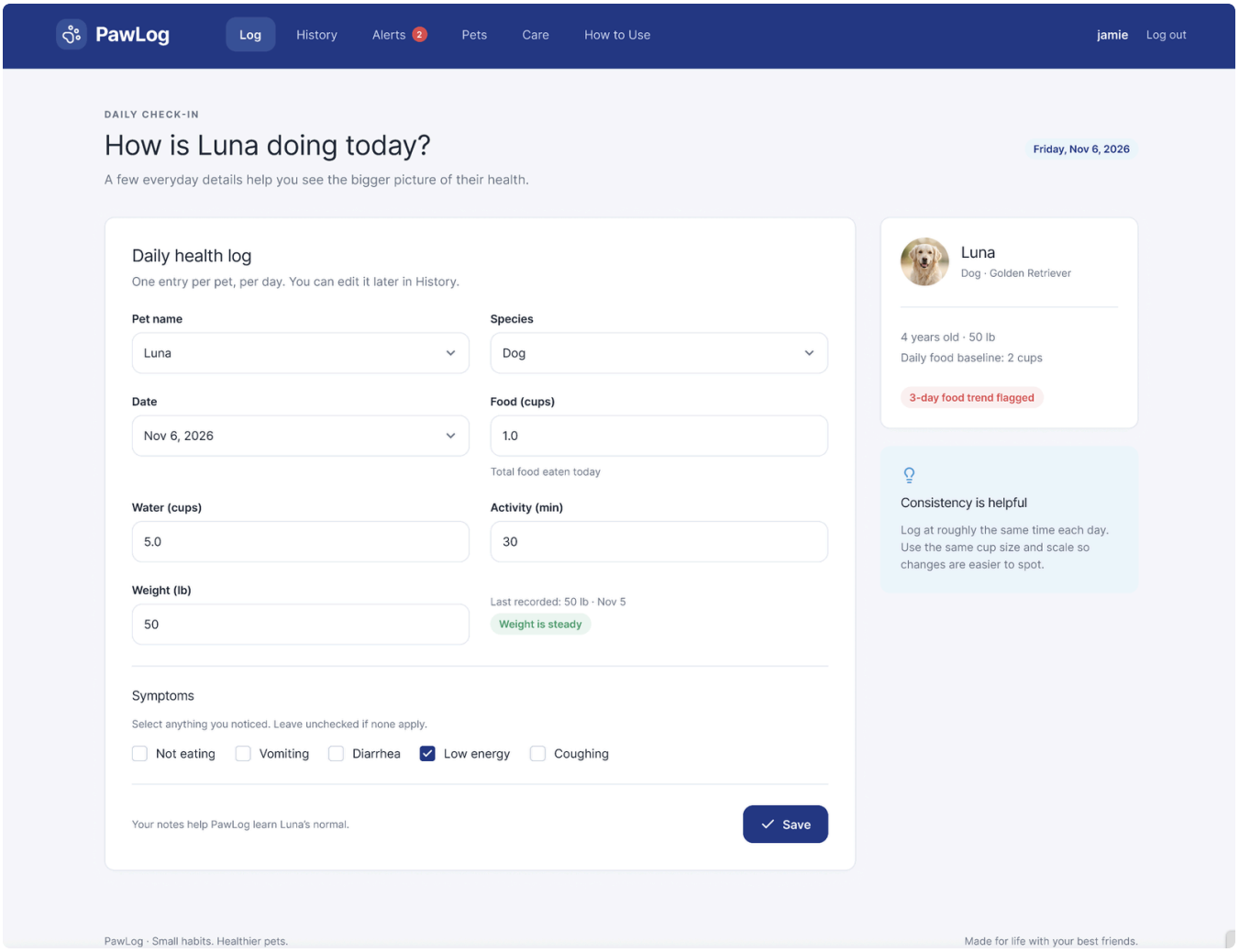
Screen	Owner	What it shows	Story
Daily Log Form	Eleah	Pet name and species, date, food (cups), water (cups), activity (minutes), weight (pounds), symptom checkboxes, Save button. Same form for editing.	1
Entry History	Eleah	List of entries, newest first. Filters for pet, date range, and symptom. Edit and Delete buttons on each entry.	1, 2
Alerts	Eleah	Flagged entries with the pet, date, what changed, and the probability (e.g. 82%). "Real problem" and "False alarm" buttons. Model accuracy and a "not veterinary advice" note.	3, 4
Pet Profiles	Bryan	List of the user's pets with name, species, breed, and age. Add, Edit, and Delete buttons.	5
Pet Profile Form	Bryan	Name, species, breed, age, and vet info fields with a Save button.	5
Care Task Form	Bryan	Pet, task type (med, vaccine, vet visit), date, repeat schedule (none, daily, weekly, yearly), Save button.	6
Care Dashboard	Bryan	Overdue tasks at the top, then upcoming tasks by date, across all pets.	7
How to Use	Shared	Short steps for logging entries, reading alerts, and managing care tasks.	All

Every screen shares a navbar with links to Log, History, Alerts, Pets, Care, How to Use, and Log out.

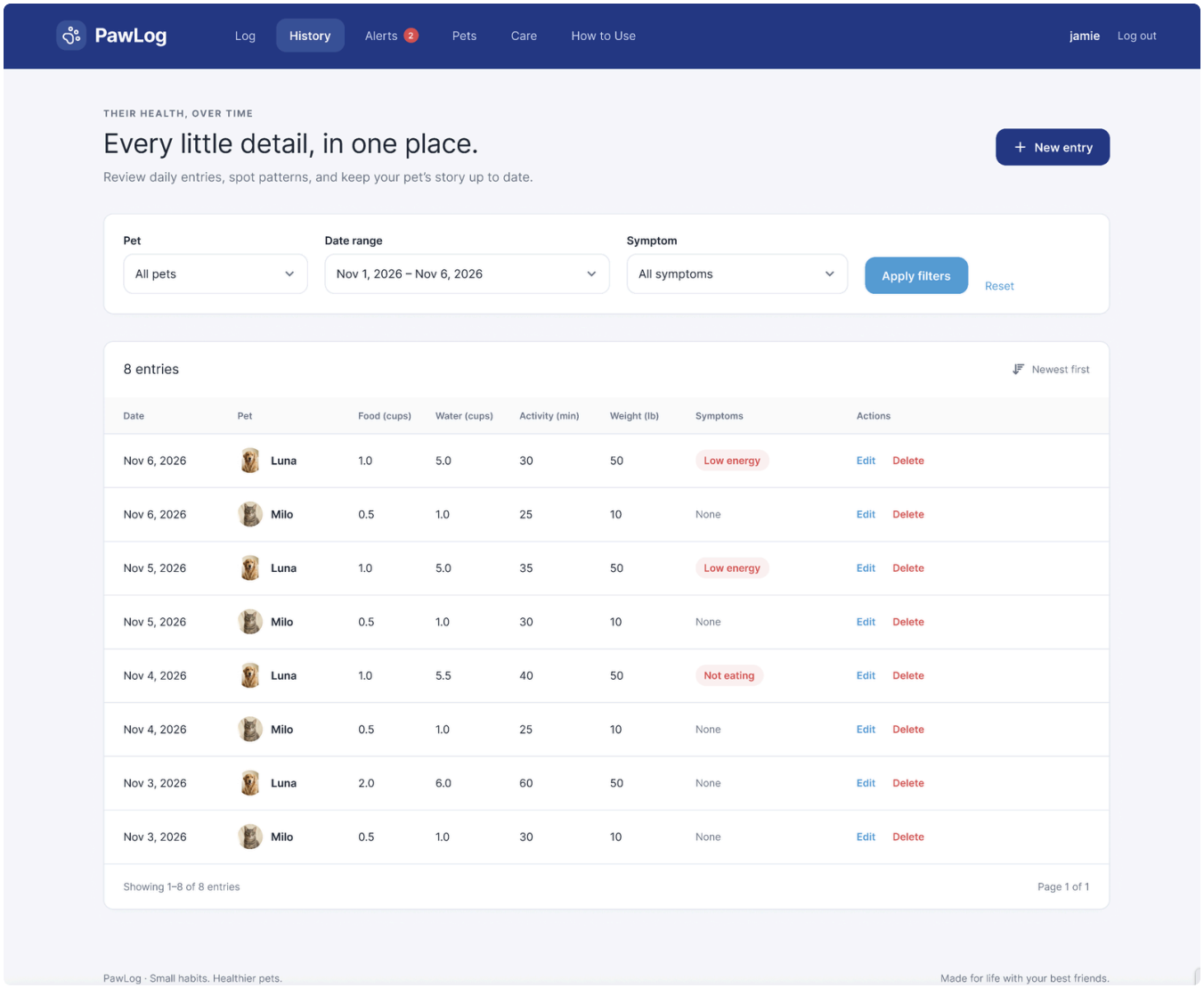
1. Login / Register



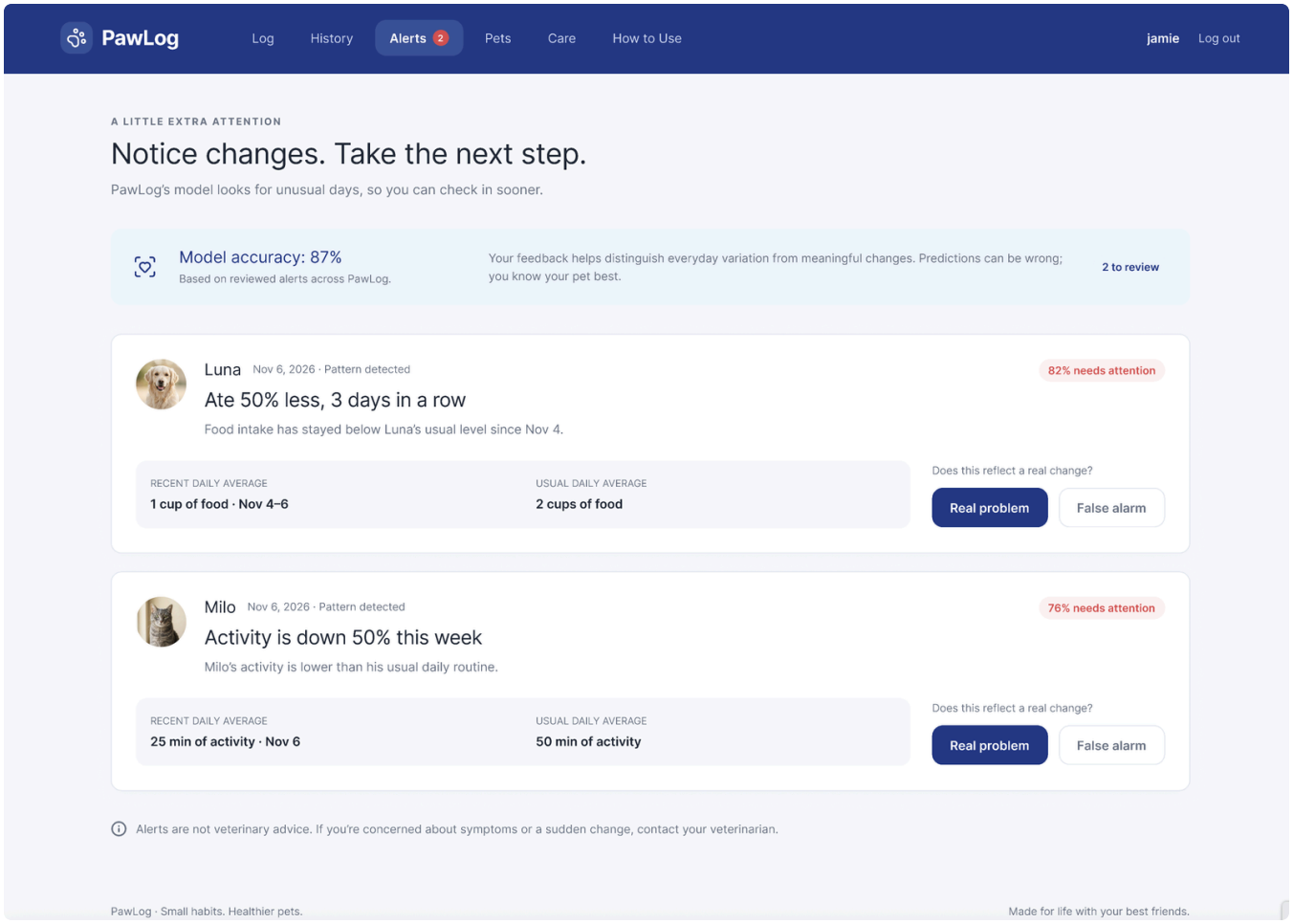
2. Daily Log Form



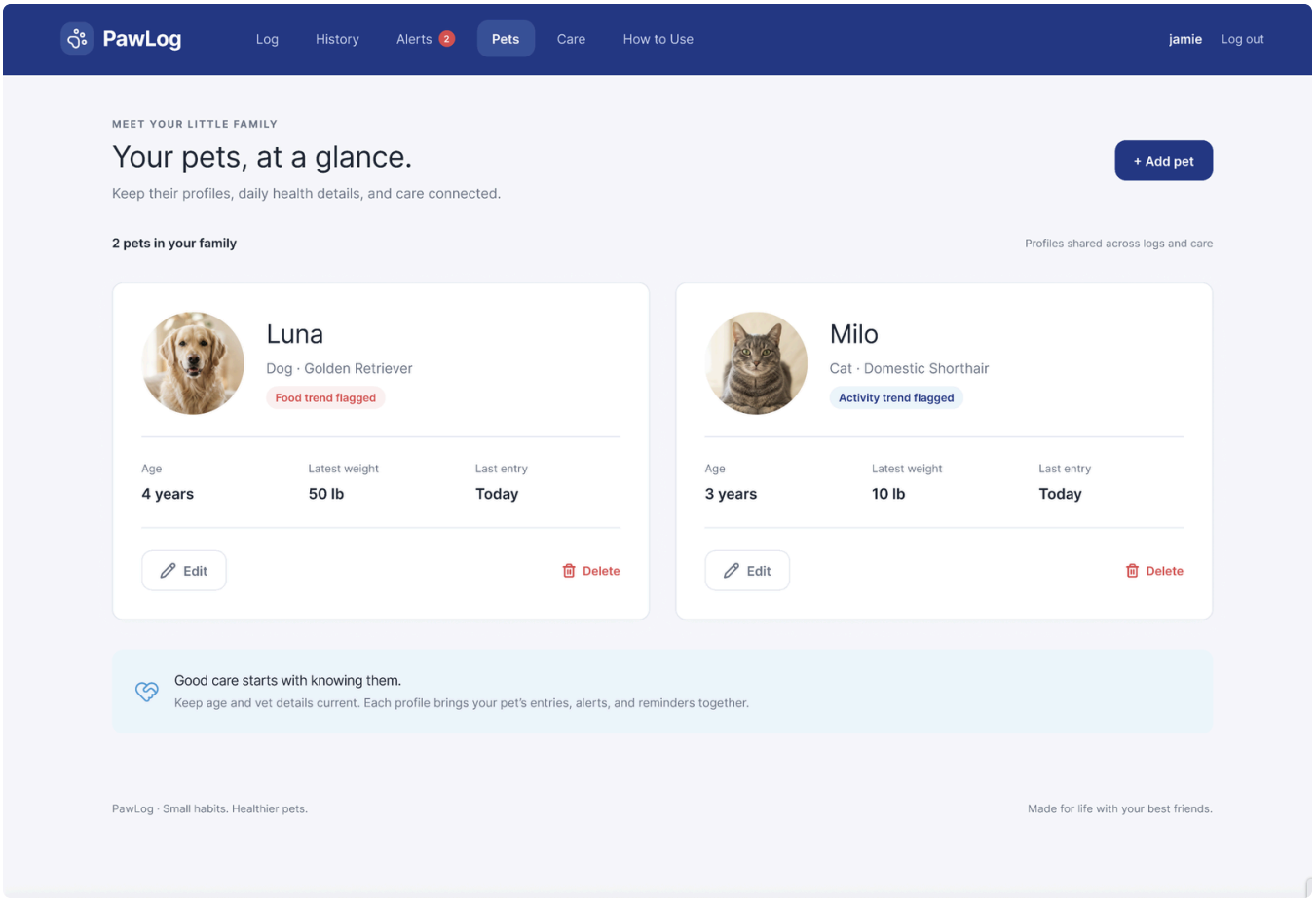
3. Entry History



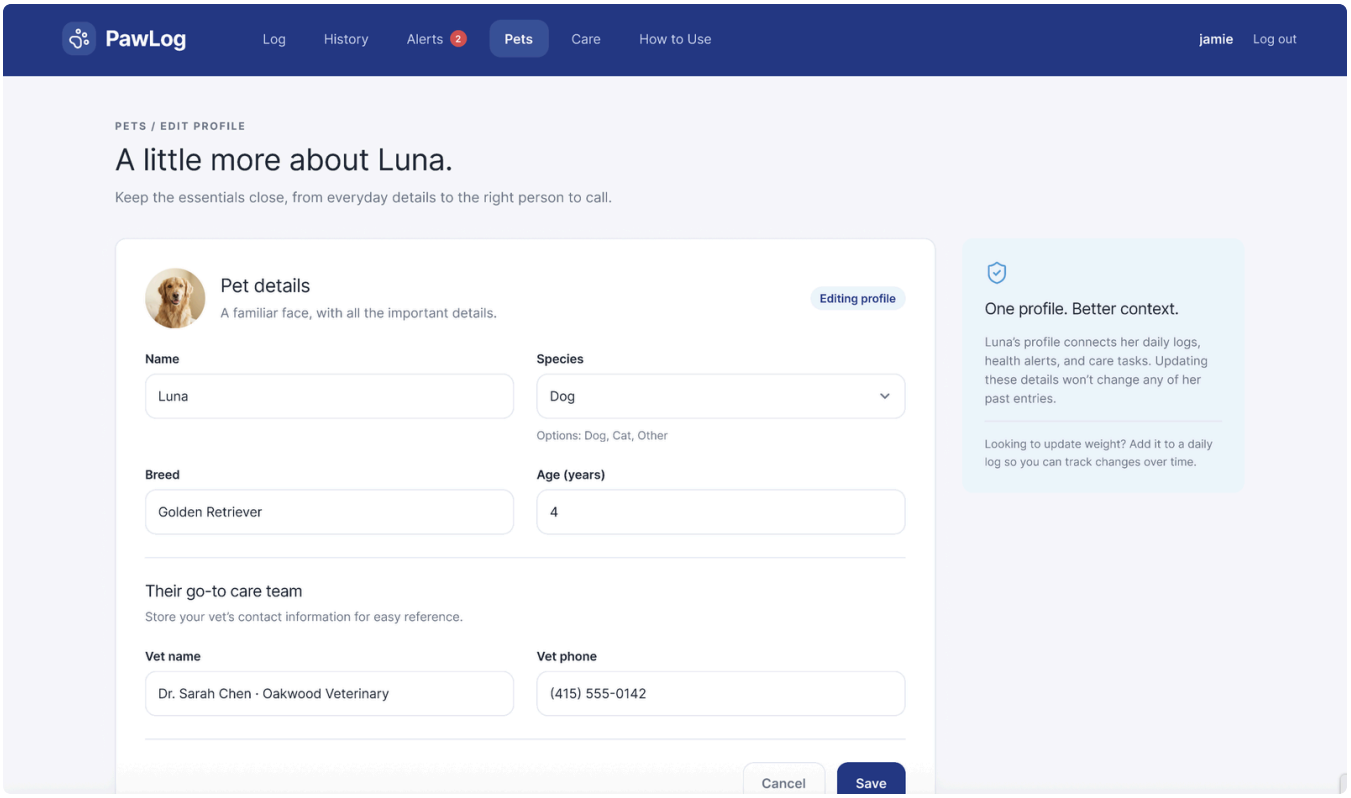
4. Alerts


PawLog
Log
History
Alerts 2
Pets
Care
How to Use
jamie
Log out
A LITTLE EXTRA ATTENTION
Notice changes. Take the next step.
PawLog's model looks for unusual days, so you can check in sooner.
Model accuracy: 87%
Based on reviewed alerts across PawLog.
Your feedback helps distinguish everyday variation from meaningful changes. Predictions can be wrong; you know your pet best.
2 to review
Luna
Nov 6, 2026 · Pattern detected
82% needs attention
Ate 50% less, 3 days in a row
Food intake has stayed below Luna's usual level since Nov 4.
RECENT DAILY AVERAGE
1 cup of food · Nov 4–6
USUAL DAILY AVERAGE
2 cups of food
Does this reflect a real change?
Real problem
False alarm
Milo
Nov 6, 2026 · Pattern detected
76% needs attention
Activity is down 50% this week
Milo's activity is lower than his usual daily routine.
RECENT DAILY AVERAGE
25 min of activity · Nov 6
USUAL DAILY AVERAGE
50 min of activity
Does this reflect a real change?
Real problem
False alarm
Alerts are not veterinary advice. If you're concerned about symptoms or a sudden change, contact your veterinarian.
PawLog · Small habits. Healthier pets.
Made for life with your best friends.

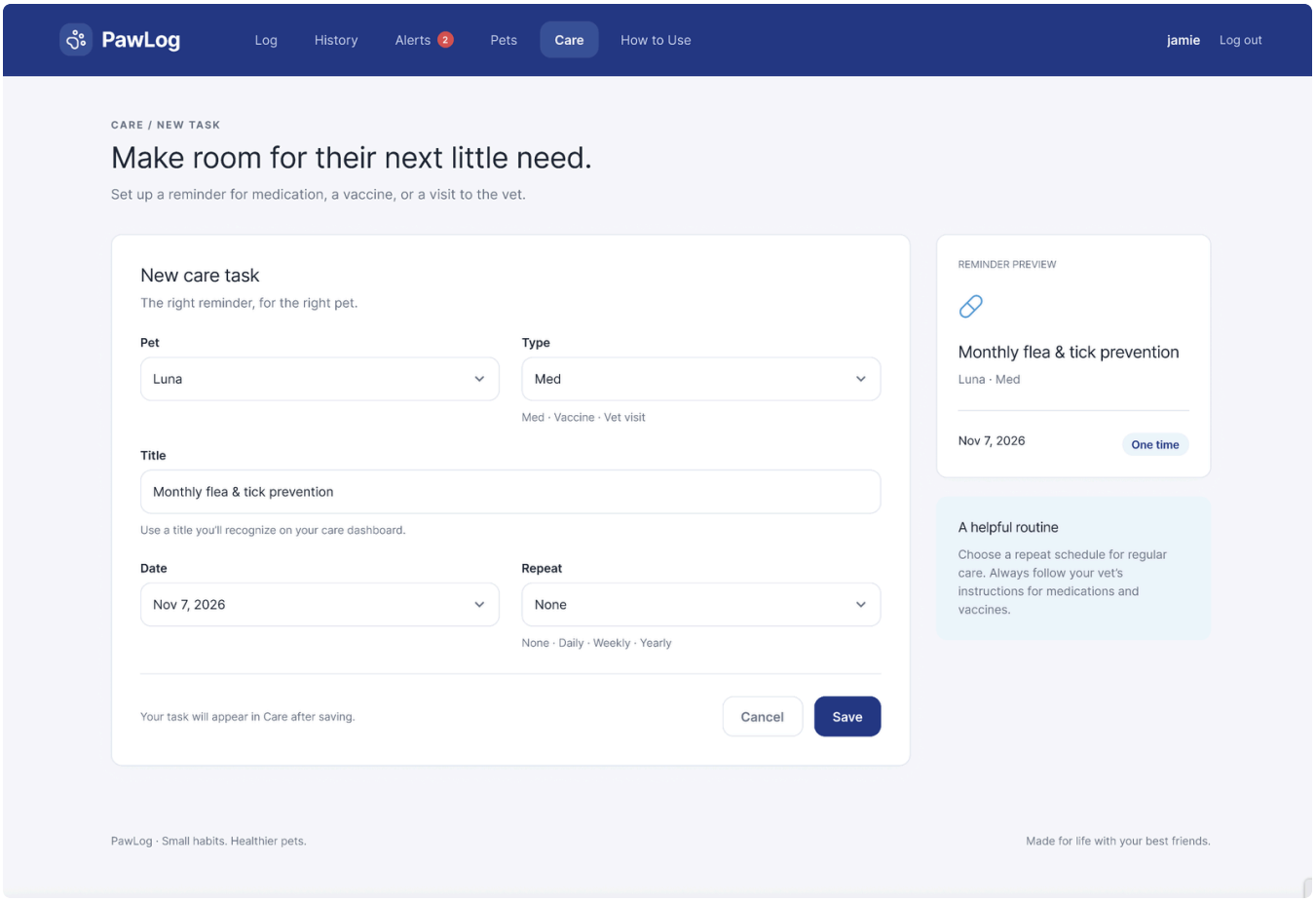
5. Pet Profiles



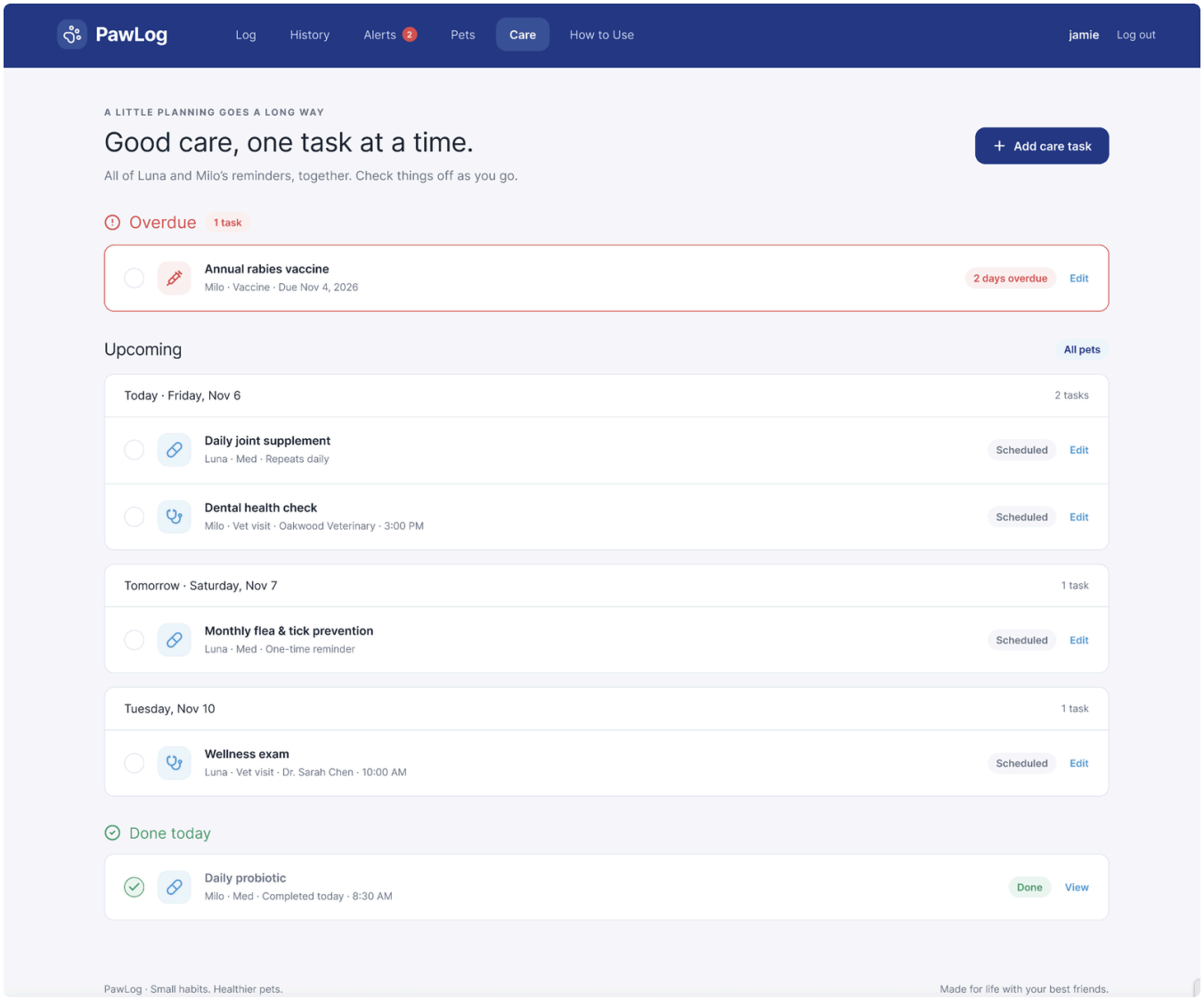
6. Pet Profile Form



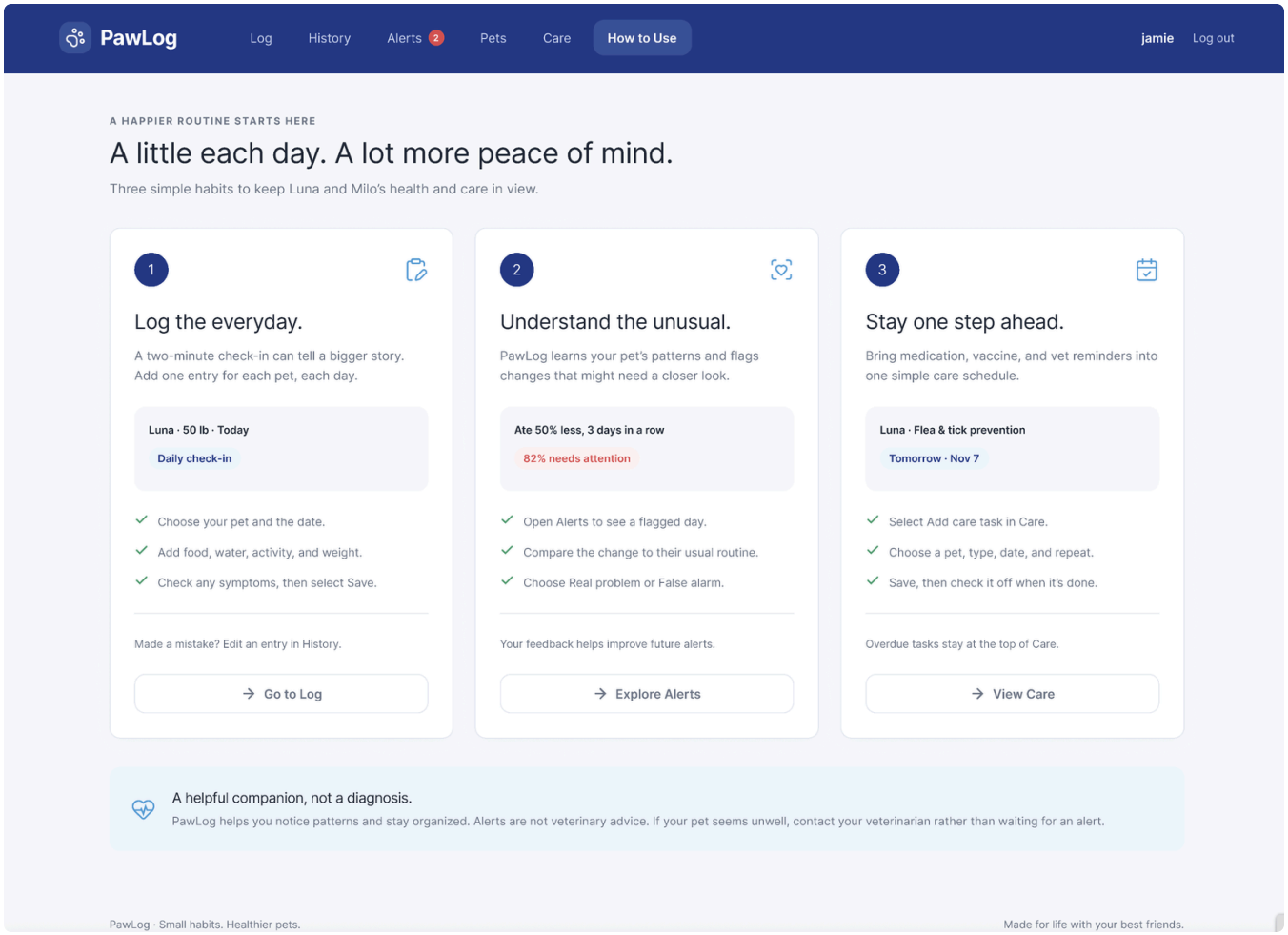
7. Care Task Form



8. Care Dashboard



9. How to Use


PawLog
Log
History
Alerts 2
Pets
Care
How to Use
jamie
Log out
A HAPPIER ROUTINE STARTS HERE
A little each day. A lot more peace of mind.
Three simple habits to keep Luna and Milo's health and care in view.
1
Log the everyday.
A two-minute check-in can tell a bigger story. Add one entry for each pet, each day.
Luna · 50 lb · Today
Daily check-in
Choose your pet and the date.
Add food, water, activity, and weight.
Check any symptoms, then select Save.
Made a mistake? Edit an entry in History.
Go to Log
2
Understand the unusual.
PawLog learns your pet's patterns and flags changes that might need a closer look.
Ate 50% less, 3 days in a row
82% needs attention
Open Alerts to see a flagged day.
Compare the change to their usual routine.
Choose Real problem or False alarm.
Your feedback helps improve future alerts.
Explore Alerts
3
Stay one step ahead.
Bring medication, vaccine, and vet reminders into one simple care schedule.
Luna · Flea & tick prevention
Tomorrow · Nov 7
Select Add care task in Care.
Choose a pet, type, date, and repeat.
Save, then check it off when it's done.
Overdue tasks stay at the top of Care.
View Care
A helpful companion, not a diagnosis.
PawLog helps you notice patterns and stay organized. Alerts are not veterinary advice. If your pet seems unwell, contact your veterinarian rather than waiting for an alert.
PawLog · Small habits. Healthier pets.
Made for life with your best friends.